LEVEL UP ACADEMY

Class 10 | CBSE | Science

ANSWER KEY & MARKING SCHEME

Pre-Board Mock 3 (CBSE pattern)

Chapter / Topic: Full syllabus (Chapters 1–13)

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SECTION A

Q1. Which of the following is NOT a redox reaction? **[1]**

Answer: (d) $BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + 2HCl$. It is a double displacement reaction; nothing is oxidised or reduced.

Q2. Which of these salts does NOT contain water of crystallisation? **[1]**

Answer: (b) Baking soda ($NaHCO_3$)

Q3. An aqueous solution has a pH of 0. It is: **[1]**

Answer: (a) strongly acidic

Q4. Which property of metals makes them suitable for making bells? **[1]**

Answer: (d) Sonority

Q5. How many covalent bonds are there in a molecule of methane? **[1]**

Answer: (c) 4

Q6. The two carbon atoms in ethyne are joined by: **[1]**

Answer: (a) a triple bond

Q7. In a plant cell, photosynthesis takes place in the: **[1]**

Answer: (b) chloroplasts

Q8. Lymph is: **[1]**

Answer: (a) a colourless fluid that also carries digested and absorbed fat

Q9. Iodine is needed by the body to make: **[1]**

Answer: (d) thyroxin

Q10. Which of the following involves sexual reproduction? **[1]**

Answer: (b) Pollination followed by fertilisation in mustard

Q11. Two heterozygous tall pea plants (Tt × Tt) are crossed. What fraction of the offspring will be tall? **[1]**

Answer: (c) 3/4 (TT, Tt, Tt are tall; tt is short)

Q12. Which of these is an artificial ecosystem? **[1]**

Answer: (b) Aquarium

Q13. Which mirror do dentists use to see enlarged images of teeth? **[1]**

Answer: (d) Concave mirror

Q14. 1 dioptre is the power of a lens whose focal length is: **[1]**

Answer: (c) 1 m

Q15. The potential difference across a resistor is measured with a: **[1]**

Answer: (c) voltmeter connected in parallel

Q16. The direction of the magnetic field around a straight current-carrying conductor is given by: **[1]**

Answer: (a) the right-hand thumb rule

Q17. **Assertion (A):** Distilled water does not conduct electricity well, but tap water does. **Reason (R):** Tap water contains dissolved salts, which provide… **[1]**

Answer: (a)

Q18. **Assertion (A):** Human beings have a four-chambered heart. **Reason (R):** Fish have a two-chambered heart. **[1]**

Answer: (b). Both are true, but R does not explain A.

Q19. **Assertion (A):** Concave mirrors are used in solar furnaces. **Reason (R):** A concave mirror concentrates sunlight at its focus. **[1]**

Answer: (a)

Q20. **Assertion (A):** Diamond is a good conductor of electricity. **Reason (R):** Graphite is a good conductor of electricity. **[1]**

Answer: (d). Diamond is a poor conductor; graphite conducts.

SECTION B

Q21. Hydrogen sulphide gas is passed through copper sulphate solution and a black precipitate forms. Write the balanced equation and name the type of… **[2]**

Answer: $CuSO_4 + H_2S \rightarrow CuS + H_2SO_4$. The black precipitate is copper sulphide. This is a double displacement (precipitation) reaction.

- Balanced equation: 1
- Type: 1

Q22. Give two differences between the chemical properties of metals and non-metals. **[2]**

Answer: Metals form basic oxides, while non-metals form acidic or neutral oxides. Many metals displace hydrogen from dilute acids, while non-metals do not. Metals lose electrons to form positive ions, while non-metals gain electrons to form negative ions.

- Any two differences: 1 + 1

OR, answer: (a) $NaHCO_3$ (b) $Na_2CO_3 \cdot 10H_2O$ (c) $CaOCl_2$ (d) $CaSO_4 \cdot ½H_2O$

- ½ each

Q23. Name the end products of the digestion of (a) carbohydrates, (b) proteins, (c) fats. **[2]**

Answer: (a) Glucose (b) Amino acids (c) Fatty acids and glycerol

- All three: 2 (two correct: 1)

Q24. Why are offspring produced by asexual reproduction almost identical to their parent? **[2]**

Answer: They come from a single parent, and its DNA is copied into them. Only small errors in DNA copying cause slight variations, so they are almost identical.

- Single parent, DNA copied: 1
- Only minor copying errors: 1

OR, answer: If the environment changes (e.g. temperature rises), some variants may be better suited to survive. The species then continues through these individuals, instead of being wiped out.

- Changing environment: 1
- Some variants survive: 1

Q25. A convex lens of focal length 50 cm is placed in contact with a concave lens of focal length 20 cm. Find the power of the combination. Is it… **[2]**

Answer: $P_1 = 1/0.5 = +2$ D. $P_2 = 1/(–0.2) = –5$ D. $P = +2 – 5 = –3$ D. Since the power is negative, the combination is diverging.

- Each power / sum: 1
- Diverging: 1

Q26. Bulbs rated 60 W and 100 W (both for 220 V) are connected in series across 220 V. Which one glows brighter? Explain. **[2]**

Answer: The 60 W bulb. Its resistance ($220^2/60 \approx 807\ \Omega$) is higher than that of the 100 W bulb (484 Ω). In series the same current flows through both, so $P = I^2R$ is greater for the 60 W bulb.

- 60 W bulb: 1
- Same current, higher R → more power: 1

OR, answer: In series the same current flows through each resistor, while in parallel the same potential difference is across each. In series the total resistance is the sum (larger than any one), while in parallel 1/R = sum of 1/R's (smaller than the smallest). In series, if one component fails the whole circuit breaks; in parallel the others keep working.

- Any two: 1 + 1

SECTION C

Q27. A compound X reacts with sodium to give hydrogen gas. When X is heated with excess concentrated sulphuric acid at 443 K, it gives an unsaturated… **[3]**

Answer: X is ethanol (C_2H_5OH) and Y is ethene (C_2H_4). $2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$. $C_2H_5OH \rightarrow C_2H_4 + H_2O$ (dehydration by hot concentrated H_2SO_4).

- X and Y: 1
- Each equation: 1 + 1

Q28. Write balanced equations for the reactions of: **[3]**

(i) $2NaOH + Zn \rightarrow Na_2ZnO_2 + H_2$ [1]
(ii) $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$ [1]
(iii) $2Cu + O_2 \rightarrow 2CuO$ [1]

Q29. How do plants respond to stimuli without a nervous system? Explain with the examples of the touch-me-not plant and a climbing tendril. **[3]**

Answer: Plants use electrical-chemical signals and changes in the water content of cells, and plant hormones that cause uneven growth. Touch-me-not: when touched, the leaves fold because cells change shape by changing their water content; no growth is involved. Tendril: the part touching the support grows more slowly than the part away from it, so the tendril coils around the support; this is a growth movement.

- General mechanism: 1
- Touch-me-not: 1
- Tendril: 1

Q30. Give the function of each gland: (i) pituitary, (ii) thyroid, (iii) adrenal. **[3]**

Answer: (i) Pituitary: secretes growth hormone, which regulates growth; deficiency in childhood causes dwarfism. (ii) Thyroid: secretes thyroxin, which regulates the metabolism of carbohydrates, proteins and fats. (iii) Adrenal: secretes adrenaline, which prepares the body for emergencies (faster heartbeat and breathing).

- Each: 1

OR, answer: Iodine is needed by the thyroid gland to make thyroxin, which regulates the metabolism of carbohydrates, proteins and fats. A lack of iodine can cause goitre (a swollen neck). Iodised salt provides enough iodine in the diet.

- Iodine for thyroxin: 1
- Thyroid gland: 1
- Goitre: 1

Q31. State the 10 per cent law. If producers in a food chain trap 1,00,000 J of energy, how much energy reaches the fourth trophic level? Why are top… **[3]**

Answer: Only about 10% of the energy at one trophic level is passed on to the next. Energy at levels 1 → 2 → 3 → 4: 1,00,000 → 10,000 → 1,000 → 100 J. Very little energy is left at the top, so it can support only a few top carnivores.

- Law: 1
- 100 J: 1
- Reason: 1

Q32. A 10 cm tall object is placed 20 cm from a convex lens of focal length 30 cm. Find the position, size and nature of the image. **[3]**

Answer: u = –20 cm, f = +30 cm. 1/v = 1/f + 1/u = 1/30 – 1/20 = –1/60, so v = –60 cm (on the same side as the object). m = v/u = (–60)/(–20) = +3, so h′ = 30 cm. The image is virtual, erect and magnified (30 cm tall).

- v = –60 cm: 1
- Size 30 cm: 1
- Nature: 1

OR, answer: f = –20 cm, v = –60 cm. 1/u = 1/f – 1/v = –1/20 + 1/60 = –2/60, so u = –30 cm. m = –v/u = –(–60)/(–30) = –2. The image is real, inverted and twice the size.

- u = –30 cm: 1
- m = –2: 1
- Nature: 1

Q33. An electric iron uses 840 W at its highest setting and 360 W at its lowest, on a 220 V supply. Find the current and the resistance in each case. **[3]**

Answer: Highest: I = 840/220 ≈ 3.82 A, R = $220^2/840$ ≈ 57.6 Ω. Lowest: I = 360/220 ≈ 1.64 A, R = $220^2/360$ ≈ 134.4 Ω.

- Currents: 1½
- Resistances: 1½

SECTION D

Q34. Answer the following: **[5]**

(i) Both react with acids to give a salt, water and carbon dioxide. $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$. $NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$. Test: pass the gas through lime water, which turns milky: $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$. [3]

- Each equation: 1 + 1
- Lime water test: 1

(ii) Curd is acidic; it reacts with copper or brass to form harmful (poisonous) compounds. [1]

(iii) A mixture of several indicators that shows different colours at different pH values (different concentrations of H^+ ions). [1]

OR, answer: (a) Combustion: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + heat and light. Oxidation: ethanol is oxidised to ethanoic acid by alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$. Addition (hydrogenation): $C_2H_4 + H_2 \rightarrow C_2H_6$, with a nickel catalyst. Substitution: $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$, in sunlight. (b) They have a higher proportion of carbon, which does not burn completely in air, leaving unburnt carbon as soot.

- (a) 1 each
- (b) 1

Q35. Answer the following: **[5]**

(i) Pollen from the anther reaches the stigma (pollination). A pollen tube grows from the pollen grain through the style to the ovule and carries the male germ cell to the female germ cell (egg), and they fuse to form a zygote (fertilisation). The zygote divides to form an embryo, the ovule develops a tough coat and becomes the seed, and the ovary grows into the fruit. [3]

- Pollination: 1
- Pollen tube and fertilisation: 1
- Seed and fruit formation: 1

(ii) Self-pollination: pollen reaches the stigma of the same flower (or another flower of the same plant), needs no outside agent, and gives little variation. Cross-pollination: pollen reaches a flower on another plant of the same species, through agents such as wind, water or insects, and gives more variation. [2]

- Any two differences: 1 + 1

OR, answer: (a) The liver secretes bile, which makes the food in the small intestine alkaline so that pancreatic enzymes can act, and breaks fats into small globules (emulsification). The pancreas secretes pancreatic juice, containing trypsin (digests proteins) and lipase (digests emulsified fats). (b)

Villi are finger-like projections that greatly increase the surface area of the small intestine, and they are richly supplied with blood vessels that carry the absorbed food to every cell.

- (a) liver 1½, pancreas 1½
- (b) surface area 1, blood supply 1

Q36. Answer the following: **[5]**

(i) A ray entering the slab obliquely bends towards the normal (air to glass: denser medium). When it comes out, it bends away from the normal (glass to air). Because the two faces are parallel, the emergent ray is parallel to the incident ray but shifted sideways. This perpendicular shift between the incident and emergent rays is called the lateral displacement. [3]

- Bending at the first face: 1
- Bending at the second face / emergent ray parallel: 1
- Lateral displacement: 1

(ii) $v = c/n = (3 \times 10^8)/1.33 \approx 2.26 \times 10^8$ m/s [2]

- Formula: 1
- Answer: 1

OR, answer: (a) Dispersion is the splitting of white light into its seven colours, because different colours bend by different amounts. A rainbow forms when sunlight falls on tiny water droplets after rain. Each droplet refracts and disperses the light, reflects it internally, and refracts it again as it leaves. The rainbow is seen opposite the Sun. (b) Red light has the longest wavelength and is scattered least by fog and smoke, so it can be seen clearly from far away.

- (a) dispersion 1, rainbow steps 2
- (b) least scattered 1, visible from far 1

SECTION E

Q37. Doctors use Plaster of Paris to support fractured bones in the right position. It is a white powder made by heating gypsum carefully at 373 K. When… **[4]**

(i) Calcium sulphate hemihydrate, $CaSO_4 \cdot ½H_2O$. [1]

(ii) By heating gypsum ($CaSO_4 \cdot 2H_2O$) at 373 K: $CaSO_4 \cdot 2H_2O \rightarrow CaSO_4 \cdot ½H_2O + 1½H_2O$. [1]

(iii) If it absorbs moisture, it slowly changes into gypsum and becomes hard, losing its setting property. Other uses (any one): making toys and decorative items; making surfaces smooth. [2]

- Sets with moisture: 1
- Another use: 1

Q38. Mendel crossed pea plants that produced yellow seeds (YY) with plants that produced green seeds (yy). All the F_1 seeds were yellow. When the F_1… **[4]**

(i) Yellow. [1]

(ii) About 3 : 1 (6022 ÷ 2001 ≈ 3). [1]

(iii) YY and Yy. Of every 3 yellow seeds, 1 is YY and 2 are Yy (the F_2 ratio is 1 YY : 2 Yy : 1 yy), so 1/3 of the yellow seeds are pure-breeding. [2]

- YY and Yy: 1
- 1/3: 1

Q39. In scrapyards, large electromagnets hanging from cranes lift heavy pieces of scrap iron. An electromagnet is a solenoid with a soft iron core. When… **[4]**

(i) A magnet made by passing current through a solenoid wound around a soft iron core. It is magnetic only while the current flows. [1]

(ii) Soft iron becomes strongly magnetised when current flows and loses its magnetism as soon as the current is switched off. [1]

(iii) Increase the current, or increase the number of turns in the coil. When the current is switched off, the core loses its magnetism and the scrap drops. [2]

- Two ways: 1
- Scrap drops: 1